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Integration of Isotopic Metrics with CO_2 Analyzers and Robust Modeling for Effective Reservoir Monitoring of a Field-Scale CO_2 Flooding Project

Mohamad Yousef Alklih, Rashed Alwaeel, Mohamed Mahmoud Abdelrahman, Karem Khan, Aamna Al Tenaiji, Tamer Koksalan, and Syed Ajmal Haider, ADNOC Onshore, Abu Dhabi, United Arab Emirates

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Abstract

CO_2-EOR is a critical technique for maximizing oil extraction from reservoirs in Abu Dhabi. Evaluating sweep efficiency and identifying breakthrough times are essential for optimizing existing and future CO_2 projects. This study focuses on utilizing CO_2 carbon isotope composition and molar concentrations along with model calibration to monitor sweep efficiency in production wells of an Onshore field. The goal is to optimize the project by adjusting the injection and future producer well placements, leveraging the distinct isotopic signature and robust modeling techniques.

CO_2 is injected through specific horizontal injectors in the northern region of the field and periodic analyses of these components' composition and isotopes in oil and water producing wells indicate the degree of breakthrough. The approach integrated all the available data from tracers, isotopes and CO_2 analyzers in order to calibrate the model. CO_2 is mostly flowing in the upper layer of the reservoir. An assessment of multiple options following an uncertainty analysis workflow is conducted; to tackle reservoir challenges. The approach also worked on increasing the permeability contrast in the best rock type to facilitate the gas flow in the upper layer, emulating possible fractures in the model.

Findings from CO_2 molar and isotope measurements in both injector and producer wells show that injected CO_2 tends to breakthrough preferentially in certain regions, indicating varied reservoir quality across the field. Isotopes tend to indicate CO_2 breakthrough sooner than the CO_2 molar concentration, demonstrating the sensitivity of isotopic metrics in detecting changes in producer wells, a finding which has also been observed clearly in one of the four wells. Data also recommended the need to convert the existing CO_2 continuous injectors into WAG as a mitigation plan to the increasing trends in breakthrough volumes.

This study presents a novel approach to monitoring CO_2 flood advancement in oil reservoirs by integrating isotopic, molar concentration data and model calibration. The use of a distinct isotopic signature for injected CO_2 provides a cost-effective and accurate method for evaluating sweep efficiency. The methodology's ability to detect changes in producer wells and the demonstrated efficiency of CO_2 WAG over continuous injection highlight the potential for optimizing EOR projects, along with offering valuable insights for future reservoir management and EOR strategies.

Introduction

CO_2-Enhanced Oil Recovery (CO_2-EOR) has emerged as a critical technique for maximizing hydrocarbon extraction from mature reservoirs in Abu Dhabi, providing the dual benefit of improving oil recovery while advancing the nation's carbon management and sustainability goals (Al-Mutairi and Graves 2008; Almulla and Al Hosani 2021). By injecting CO_2 into depleted or under-pressured reservoirs, operators can mobilize residual oil through mechanisms such as oil swelling, viscosity reduction, and miscibility development, which collectively enhance displacement efficiency (Lake et al. 2014). However, the success of CO_2-EOR projects depends heavily on precise monitoring of injected CO_2 to evaluate sweep efficiency, identify breakthrough times, and guide real-time operational adjustments (Han et al. 2010).

This study addresses these challenges by employing an integrated monitoring methodology that leverages CO_2 carbon isotope compositions, molar concentration measurements, and advanced model calibration to track CO_2 migration and distribution across production wells. Isotopic analysis provides a particularly sensitive and cost-effective tracer for monitoring CO_2 movement, as the injected gas carries a distinct isotopic "fingerprint" that can be measured in produced fluids to map breakthrough and reservoir sweep (Benson and Cole 2008; Pearce et al. 2013). By detecting subtle isotopic variations, operators can gain early insight into the extent of CO_2 mixing with reservoir fluids, identify unswept or bypassed zones, and refine dynamic models to more accurately predict fluid movement under ongoing injection scenarios.

The ultimate goal is to optimize project performance through data-driven adjustments to injection rates, volumes, and future producer well placements, enabling more efficient displacement of oil and mitigating early CO_2 breakthrough that reduces recovery efficiency (Han et al. 2010; Almulla and Al Hosani 2021). Yet, monitoring CO_2-EOR projects presents several persistent challenges. Isotopic signals may be masked or distorted by reservoir heterogeneities including variations in permeability, natural fractures, and residual oil saturation that complicate interpretation of tracer data (Al-Mutairi and Graves 2008). Further, geochemical processes such as CO_2 dissolution in formation water, phase behavior shifts, and mineralization reactions with carbonate rocks can alter the isotopic signature over time, introducing uncertainty into the tracing of injected CO_2 pathways (Benson and Cole 2008).

Beyond technical interpretation issues, integrating isotopic and molar data into reservoir models requires high-frequency sampling, robust laboratory workflows, and sophisticated simulation calibration techniques to reconcile multiple data streams, all of which increase project complexity and cost (Pearce et al. 2013). Large-scale operations add logistical constraints ensuring consistent, quality-controlled sampling across numerous wells and integrating real-time analyzer data with model updates demands significant resources and coordination.

Despite these hurdles, the use of integrated monitoring frameworks grounded in isotopic analysis and dynamic modeling represents one of the most promising pathways to overcome these uncertainties and unlock the full potential of CO_2-EOR in Abu Dhabi. When executed effectively, this approach not only enhances sweep efficiency and improves ultimate recovery but also provides long-term assurance that injected CO_2 is being efficiently utilized and securely stored, reinforcing the viability of CO_2-EOR as both an economic and environmental solution for the region (Tzimas and Georgakaki 2005; Almulla and Al Hosani 2021).

Literature Review

CO_2-EOR has been extensively studied and deployed worldwide, particularly in mature oil provinces in North America and the Middle East, as a means of both boosting oil production and advancing carbon management objectives. Since the first commercial CO_2-EOR projects in the Permian Basin in the 1970s, the technique has evolved considerably, with a growing emphasis on monitoring injected CO_2 to ensure operational efficiency, early detection of breakthrough, and verification of long-term CO_2 containment (Lake et al. 2014). Globally, more than 140 CO_2-EOR projects are currently operational, with varying levels of monitoring sophistication depending on reservoir complexity and regulatory requirements (Han et al. 2010; Benson and Cole 2008).

Several studies underscore the importance of tracking CO_2 migration pathways to improve sweep efficiency and reduce early CO_2 breakthrough. Traditional methods such as production data analysis, pressure monitoring, and 4D seismic surveys have provided valuable insights but often lack the resolution to detect subtle breakthrough trends in heterogeneous reservoirs (Pearce et al. 2013). As a result, there has been growing interest in geochemical and isotopic techniques that exploit the distinct chemical and isotopic "fingerprints" of injected CO_2 to trace its movement in the subsurface (Benson and Cole 2008).

Isotopic analysis, in particular, has emerged as a powerful monitoring tool. Studies from U.S. EOR projects and European pilot sites demonstrate that stable carbon isotopes of CO_2 ($\delta^{13}C$) can provide early indicators of breakthrough in production wells, often detecting injected CO_2 weeks to months before changes in molar concentrations are observed (Han et al. 2010). These findings align with more recent Middle Eastern research, which have confirmed the higher sensitivity of isotopic measurements for breakthrough detection and sweep assessment (Almulla and Al Hosani 2021).

In addition to isotopic tracing, water-alternating-gas (WAG) injection strategies have been widely studied as a means of mitigating CO_2 channeling and improving sweep in heterogeneous reservoirs (Tzimas and Georgakaki 2005; Al-Mutairi and Graves 2008). Numerous case studies show that switching from continuous CO_2 injection to WAG reduces breakthrough volumes and enhances overall recovery by controlling gas mobility and promoting a more uniform displacement front.

Despite these advances, several monitoring challenges persist across global CO_2-EOR projects. Reservoir heterogeneities particularly in carbonate systems can cause injected CO_2 to preferentially flow through high-permeability streaks or fractures, bypassing significant oil volumes and complicating the interpretation of tracer and isotopic data (Al-Mutairi and Graves 2008). Furthermore, geochemical processes such as CO_2 dissolution, precipitation, and mineralization can alter isotopic signatures over time, requiring careful calibration to distinguish between signal distortion and true reservoir behavior (Benson and Cole 2008).

As a result, recent literature emphasizes the need for integrated monitoring frameworks that combine isotopic and molar data with reservoir simulation, tracer programs, and, where feasible, geophysical surveys. This integration enables operators to reconcile different datasets and iteratively calibrate reservoir models for better predictive performance (Pearce et al. 2013). The methodology applied in this study aligns with these global best practices, drawing on lessons from mature CO_2-EOR provinces to establish a robust, scalable monitoring approach.

CO_2 Injection Surface Facilities

Injected CO_2 originate from natural domes, captured flue gas, or commercial pipelines. CO_2 specifications typically require water content ≤50 ppmv, minimal O_2 and H_2S, and low hydrocarbon contamination. Front-end conditioning involves multi-stage compression with intercoolers and knock-out drums, followed by dehydration using triethylene glycol (TEG) or molecular sieves to ensure ultra-dry CO_2. Filtration and polishing steps remove residual particulates and trace solvents. Dense-phase CO_2 is transported via pipelines designed to remain above critical pressure (≈74 bar) and avoid the triple-point region.

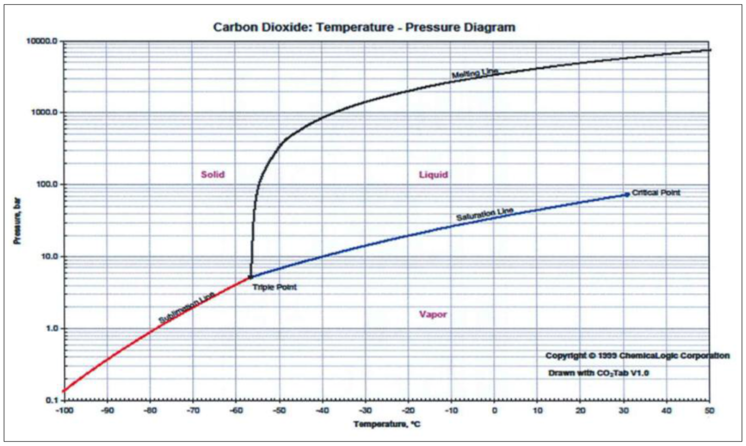

Figure 1—CO_2 Phase Diagram.

Manifolds are equipped with corrosion-resistant internals, block valves, and pig launchers/receivers. Injection lines include wellhead skids with flow control valves, chemical dosing systems (corrosion inhibitors, anti-foam agents), and instrumentation for pressure and temperature monitoring. Wells are completed using corrosion-resistant alloy (CRA) tubing or carbon steel with appropriate corrosion allowance. Elastomers compatible with CO_2, such as FKM, HNBR, and PTFE, are employed. Packers and completion hardware are designed to handle supercritical CO_2 and associated formation brines. Continuous injection of corrosion inhibitors, scale inhibitors, and anti-foam agents ensures system integrity. Methanol or MEG may be used for hydrate mitigation when water ingress is possible. Produced fluids containing high CO_2 require careful handling to prevent corrosion, foaming, and process upset. The production train includes CO_2-rated wellheads, gathering lines, and three-phase separators with CRA-lined internals. Produced gas undergoes sweetening using physical or chemical solvents, membranes, or hybrid systems. Recompression and dehydration enable CO_2 recycle to injection manifolds. Produced oil is stabilized, ensuring CO_2 does not cause excessive vapor pressure, while produced water is degassed, treated for scale control, and neutralized to maintain pH.

Operations maintain CO_2 in the dense phase, avoiding flashing across the triple point, with line heaters where necessary. Dehydration to ≤50 ppm H_2O is essential to prevent corrosion and hydrate formation. Carbon steel with corrosion allowance and chemical inhibitors is suitable for dry CO_2. CRAs, including 13Cr, 22Cr, and duplex alloys, are used in wet CO_2 or brine contact sections. Elastomers are selected for CO_2 compatibility, and coatings are applied to mitigate erosion-corrosion. High-pressure safety valves, dense-phase depressurization design, CO_2 detectors, ventilation, and process alarms are implemented. The SIS ensures safe operation during high-CO_2 events or system trips. Dispersion modeling and stack design mitigate environmental impact. Injection and production operations follow controlled ramp-up procedures to avoid phase transition and hydrate formation. Daily monitoring includes moisture content, line pressure drop, compressor performance, and corrosion indicators. Routine analysis of amine loading, heat-stable salts, foaming tendencies, and isotopic measurements ensures early detection of breakthrough and operational anomalies.

Challenges due to presence of high CO_2 include severe corrosion in wet sections, hydrate formation, amine degradation, reduced gas calorific value, oil stabilization issues, scaling in produced water, and measurement errors in metering. Effective material selection, inhibitor dosing, and monitoring are essential to manage these risks. Continuous isotopic monitoring using $\delta^{13}C$-CO_2 measurements via cavity ring-down spectroscopy (CRDS) and NDIR analyzers allows differentiation of injected versus native CO_2. Chemical tracers and routine PVT/GOR analysis complement early detection. WAG optimization, foam-assisted CO_2, polymer or gel treatments, and smart completions improve sweep efficiency and minimize channelling.

Recycle loops handle early breakthrough, selective production limits high-CO_2 wells, and separator pressure optimization keeps CO_2 dissolved in oil. On-the-fly sweetening upgrades accommodate transient CO_2 surges. Infilling injectors, pattern realignment, barrier waterfloods, and, in some cases, alternative gases maintain reservoir pressure and optimize storage.

CO_2 Molar and Isotope Measurements

CO_2 is injected through strategically placed horizontal injectors in the northern sector of the field to maximize reservoir contact and promote efficient displacement of remaining hydrocarbons. The horizontal injector configuration was selected to enhance areal coverage and minimize localized channeling, which is a common issue in conventional vertical injection schemes (Lake et al. 2014). To monitor the displacement process, periodic analyses of CO_2 composition and isotopic ratios are conducted from oil and water production wells. These measurements provide critical indicators of CO_2 breakthrough and sweep efficiency, enabling operators to better understand the dynamic interaction between injected CO_2 and in-situ fluids (Han et al. 2010; Pearce et al. 2013). Isotopic monitoring, in particular, helps identify early breakthrough and unmapped migration pathways that may not be apparent through production data alone (Benson and Cole 2008).

An integrated monitoring approach was developed, combining tracer studies, isotopic analyses, and real-time CO_2 analyzer data to calibrate the dynamic reservoir simulation model (Almulla and Al Hosani 2021). This comprehensive integration allows the model to reflect both the static geological heterogeneities and the evolving reservoir response to injection, thus providing a robust tool for forecasting CO_2 migration patterns and production behavior. Data from these calibrated models confirmed that CO_2 preferentially migrates through the upper reservoir layers, likely due to gravitational segregation and permeability stratification (Tzimas and Georgakaki 2005). This vertical migration trend can create sweep imbalance, leaving significant oil volumes in the lower zones unrecovered a recurring challenge in CO_2-EOR projects worldwide (Al-Mutairi and Graves 2008).

To address these issues, an uncertainty analysis workflow was implemented to evaluate a range of development and operational scenarios. This workflow identified key sensitivities in the system, such as permeability contrasts and injection strategies, that could influence sweep efficiency and CO_2 containment. One targeted strategy involved increasing the permeability contrast within the most favorable rock layers, effectively enhancing CO_2 movement in the upper zones while mimicking the potential effects of natural or induced fractures within the simulation model. Emulating these fracture-like pathways provides a more realistic representation of reservoir behavior, allowing for improved prediction of flow dynamics.

The calibrated model and scenario analysis guided operational recommendations, including optimal injector and producer well placement, adjustments to injection rates, and long-term CO_2 management strategies.

Isotopes have gained significant attention in enhanced oil recovery (EOR) due to their numerous advantages, including cost-effectiveness, safety, and the ability to provide detailed insights into reservoir dynamics that are often difficult to obtain through conventional monitoring methods. Isotopes are variants of atoms of the same element that have identical numbers of protons but differ in the number of neutrons, resulting in different atomic masses. These isotopes occur naturally in reservoir materials such as rocks, petroleum, natural gas, and formation water, providing intrinsic geochemical fingerprints. By introducing injected CO_2 with a distinct isotopic signature often sourced from industrial emitters such as steel mills operators can differentiate it from the native CO_2 associated with the oil, which typically has a heavier isotopic composition. This clear isotopic contrast enables precise tracking of the injected CO_2 as it migrates through the reservoir, making it possible to determine sweep efficiency and identify breakthrough levels with higher sensitivity than traditional concentration measurements. Early detection of CO_2 breakthrough

through isotopic monitoring allows for timely optimization of injection strategies, ultimately improving oil recovery and enhancing the economic and environmental performance of CO_2-EOR projects.

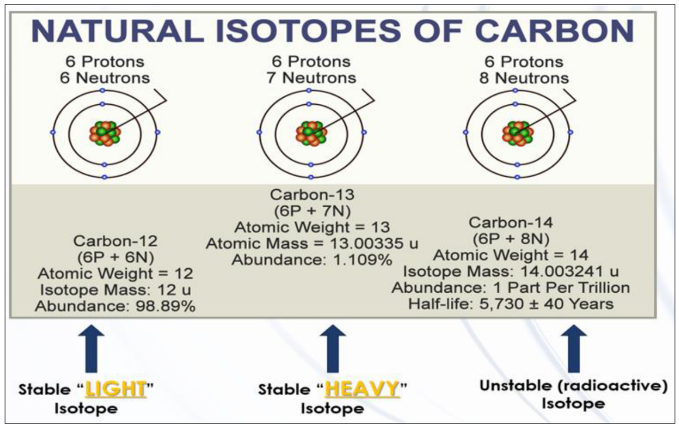

Figure 2—Isotopes (Stable and Radioactive) of Carbon in Earth.

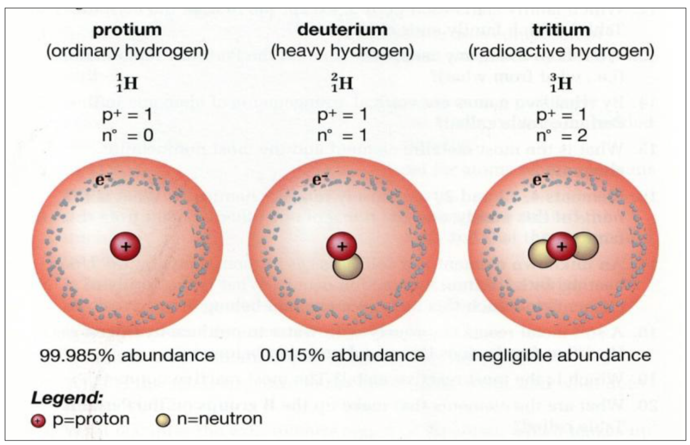

Figure 3—Stable and Radioactive Isotopes of Hydrogen on Earth.

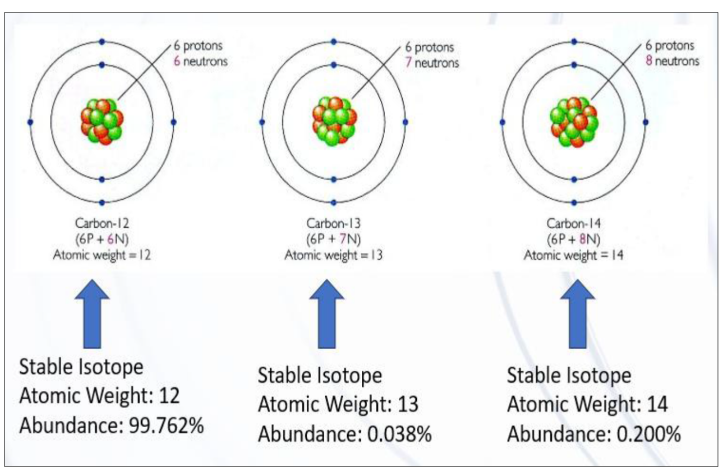

Figure 4—Stable Isotopes of Oxygen on Earth.

Gas Chromatography–Isotope Ratio Mass Spectrometry (GC-IRMS) is a key analytical technique widely used in carbon capture and storage (CCS) and Enhanced Oil Recovery (EOR) projects to accurately determine isotopic ratios such as ^13C/^12C, ^18O/^16O, and ^2H/^1H within individual compounds following chromatographic separation (Han et al. 2010; Pearce et al. 2013). The GC component vaporizes the sample and separates its constituent compounds based on their chemical properties, with each compound eluting at a characteristic retention time. Subsequently, these isolated compounds are introduced into the IRMS, where they are converted into simple gases—carbon dioxide for carbon isotopes, hydrogen gas for hydrogen isotopes, and either carbon dioxide or water for oxygen isotopes—prior to ionization and mass separation.

In the IRMS, ionized gas molecules are separated according to their mass-to-charge ratios by magnetic fields, allowing for precise measurement of the relative abundances of heavy and light isotopes (Benson and Cole 2008). The isotopic ratios are typically expressed in delta (δ) notation, which quantifies the deviation of the sample's isotope ratio from an internationally accepted standard reference, such as the Vienna Pee Dee Belemnite (VPDB) for carbon isotopes and Vienna Standard Mean Ocean Water (VSMOW) for hydrogen and oxygen isotopes (Pearce et al. 2013). Delta values, expressed in parts per thousand (‰), provide insights into fractionation processes driven by physical and chemical mechanisms such as photosynthesis, respiration, evaporation, and temperature-dependent isotope effects (Benson and Cole 2008). Notably, isotopic fractionation increases at lower temperatures because differences in the behavior of isotopes with varying masses become more pronounced under colder conditions.

The delta value (δX) is calculated using the formula:

$$\delta X = (RsampleRstandard - 1) \times 1000\delta X = (RstandardRsample - 1) \times 1000$$

where RsampleRsample and RstandardRstandard represent the ratios of heavy to light isotopes in the sample and standard, respectively (Han et al. 2010). A positive δ value indicates enrichment of the heavy isotope relative to the standard, while a negative value indicates depletion. Standards such as VPDB and VSMOW ensure accuracy, consistency, and comparability of isotopic data across different laboratories and studies (Pearce et al. 2013).

An example system consists of an Agilent GC combustion apparatus coupled to a Delta V Plus or Delta Plus Advantage mass spectrometer, optimized for δ^13C measurements of hydrocarbon gases. Gas samples may be introduced manually or via autosampler through a split/splitless injector. For pure CO_2 samples, the gas bypasses the GC and is directly ionized for mass spectrometric analysis. During IRMS, carbon ions (^12C and ^13C) are separated by their mass-to-charge ratio under a magnetic field, with heavier ^13C ions following a longer trajectory than the lighter ^12C ions (Han et al. 2010). This method enables precise tracking of CO_2 sources and migration pathways, which is critical for monitoring CO_2-EOR processes and verifying carbon storage integrity. The electric signal generated by these ions is subsequently amplified and recorded, allowing precise quantification of the 13C/12C ratio relative to the Pee Dee Belemnite standard.

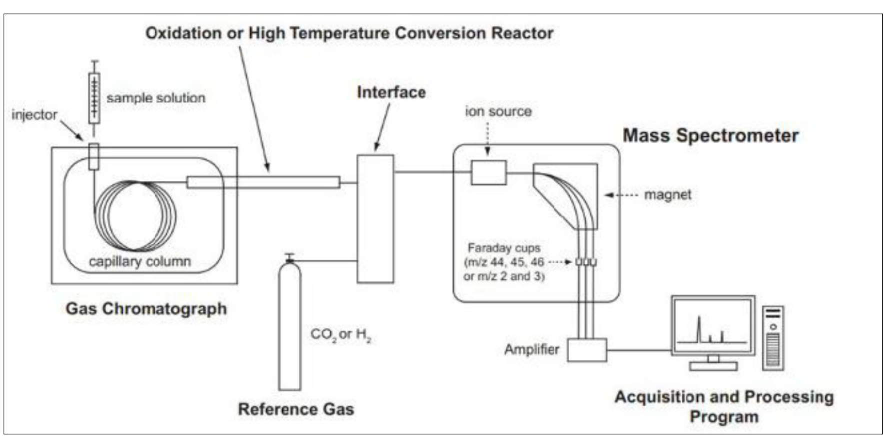

Figure 5—A simplified schematic of a GC-IRMS system that can be configured for either δ13C or δ2H measurements.

Findings from CO_2 molar and isotope measurements in both injector and producer wells show that injected CO_2 tends to break through preferentially in certain regions, highlighting varied reservoir quality across the field. Isotopic data proved especially valuable, as it tended to indicate CO_2 breakthrough sooner than molar concentration measurements a sensitivity observed particularly in one of the four monitored wells. This early-warning capability demonstrates the unique strength of isotopic metrics for detecting CO_2 movement and refining reservoir models.

Data also supported the operational recommendation to convert existing continuous CO_2 injectors to water-alternating-gas (WAG) injection. This shift is expected to mitigate increasing breakthrough volumes, improve sweep balance, and slow CO_2 channeling in high-permeability zones.

Overall, the study demonstrates a novel monitoring framework that leverages isotopic and molar concentration data alongside model calibration to track CO_2 flood advancement. The use of distinct isotopic signatures for injected CO_2 provides a cost-effective and accurate way to evaluate sweep efficiency, while the demonstrated effectiveness of WAG injection over continuous CO_2 highlights a viable path for enhancing oil recovery and managing CO_2 utilization more effectively.

CO_2 and CO_2 WAG projects are in progress in an Abu Dhabi oil field. The project features horizontal producers and injectors, supported by pilot observation wells strategically placed to monitor the movement of the flood front away from the injectors. This setup generates a substantial volume of monitoring data, which is critical for evaluating the pilot's performance and informing operational decisions.

A central component of the monitoring program is the carbon and oxygen isotope analysis of the injected and produced CO_2. By examining the distinct isotopic signatures of the CO_2, operators can effectively track the timing and extent of CO_2 breakthrough into production wells. This isotopic tracking provides early detection of injected CO_2 migration, enabling timely adjustments to injection parameters and well placements to optimize sweep efficiency.

Furthermore, combining isotope data with other reservoir monitoring tools, such as pressure and production rate measurements, enhances the understanding of reservoir heterogeneity and flow pathways. This integrated approach supports refining reservoir models, improving predictive accuracy, and guiding the transition from continuous CO_2 injection to water-alternating-gas (WAG) schemes. The WAG method aims to mitigate early CO_2 channeling by alternating CO_2 with water injection, which enhances mobility control and improves volumetric sweep efficiency.

Ongoing analysis of the monitoring data continues to inform project optimization strategies, with the goal of maximizing oil recovery while ensuring effective utilization and storage of CO_2 within the reservoir.

The figure below presents time-series plots of carbon dioxide (CO_2) isotope data for five producer wells (Wells E, F, G, H, and I) and the CO_2 injector well (Injector) spanning approximately four years, from September 2019 to November 2023 (Koksalan et al., 2024). The CO_2 injector well displays a distinct carbon-13 isotope signature, characterized by notably light, negative $\delta^{13}C$ values (depleted in ^{13}C). Over the course of the project, the $\delta^{13}C$ value of the injected CO_2 has become progressively lighter, reflecting multiple injection campaigns—initially between December 2019 and July 2020, and most recently in May 2022. Because the injected CO_2 originated from different streams at a steel manufacturing plant, the $\delta^{13}C$ values fluctuated rather than remaining constant, ranging from –42‰ to –35‰ during the injection period.

In contrast, $\delta^{13}C$ values measured in the producer well CO_2 samples have remained relatively stable, showing only minor variations over the same four-year timeframe and not trending toward the $\delta^{13}C$ signature of the injected CO_2 (Koksalan et al., 2024). This stability indicates that the injected CO_2 has not yet migrated to the production wells, as there is no observable isotopic shift toward the depleted $\delta^{13}C$ range associated with the injected gas. The isotope plots therefore provide clear evidence that CO_2 breakthrough has not occurred in the monitored producer wells to date, underscoring the value of $\delta^{13}C$ tracking in assessing flood front progression and breakthrough timing (Koksalan et al., 2024).

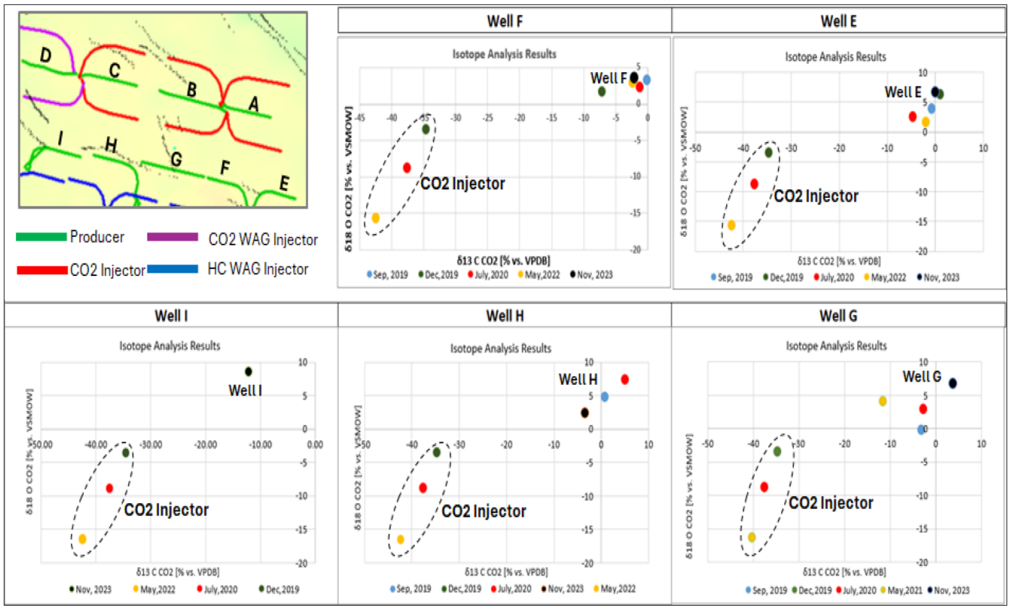

Figure 6—CO_2-WAG injectors not affected by CO_2 injection (Koksalan et al., 2024).

The figure below illustrates isotope data collected from four wells monitored for the partial and complete effects of CO_2 injection over a four-year period, with injection activities commencing in 2016 (Koksalan et al., 2024). Wells A, B, and C display a uniform and anticipated response to CO_2 injection, reflecting typical reservoir behavior under miscible flood conditions. Following the initiation of injection, Wells A and C achieved full saturation with injected CO_2 by May 2021, while Well B reached full saturation significantly later, in November 2023—more than six years after injection began. This saturation is evidenced by the convergence of $\delta^{13}C$ values between the injected CO_2 and the CO_2 produced from these wells, after which the isotopic signatures remained identical in subsequent measurements, confirming sustained CO_2 breakthrough and full reservoir sweep at these locations (Koksalan et al., 2024).

A closer examination of the isotope plots underscores that the timing and degree of injected CO_2 impact vary significantly between wells, a disparity largely attributable to the reservoir's heterogeneous nature. Wells intersecting more permeable streaks or layers register the influence of injected CO_2 earlier, whereas those drilled through less permeable zones experience delayed response (Koksalan et al., 2024).

In contrast, Well D displays an anomalous $\delta^{13}C$ trend during the monitoring period. Initially, the reservoir $\delta^{13}C$ value measured in July 2020 was –5‰, while the injected CO_2 exhibited a $\delta^{13}C$ value of –38‰. By May 2021, Well D's reservoir $\delta^{13}C$ value shifted toward the injected CO_2 signature, reaching 28%, suggesting partial breakthrough (Koksalan et al., 2024). However, by October 2022, the $\delta^{13}C$ value abruptly spiked to +8‰ before returning to –12‰ by November 2023, approximately 13 months later. This unusual fluctuation is attributed to the CO_2 Water-Alternating-Gas (WAG) injection cycles associated with the injector wells. When CO_2 slugs arrive at the producer well, the $\delta^{13}C$ values trend toward the lighter injected CO_2 signature, indicating breakthrough. Conversely, during water injection phases of the WAG cycle, the $\delta^{13}C$ signature of produced CO_2 temporarily diverges, becoming isotopically "heavier" due to dilution and mixing effects with reservoir fluids. This cyclical isotopic behavior provides direct evidence of alternating CO_2 and water fronts moving through the reservoir and highlights the utility of isotope monitoring for interpreting the complex dynamics of WAG operations (Koksalan et al., 2024).

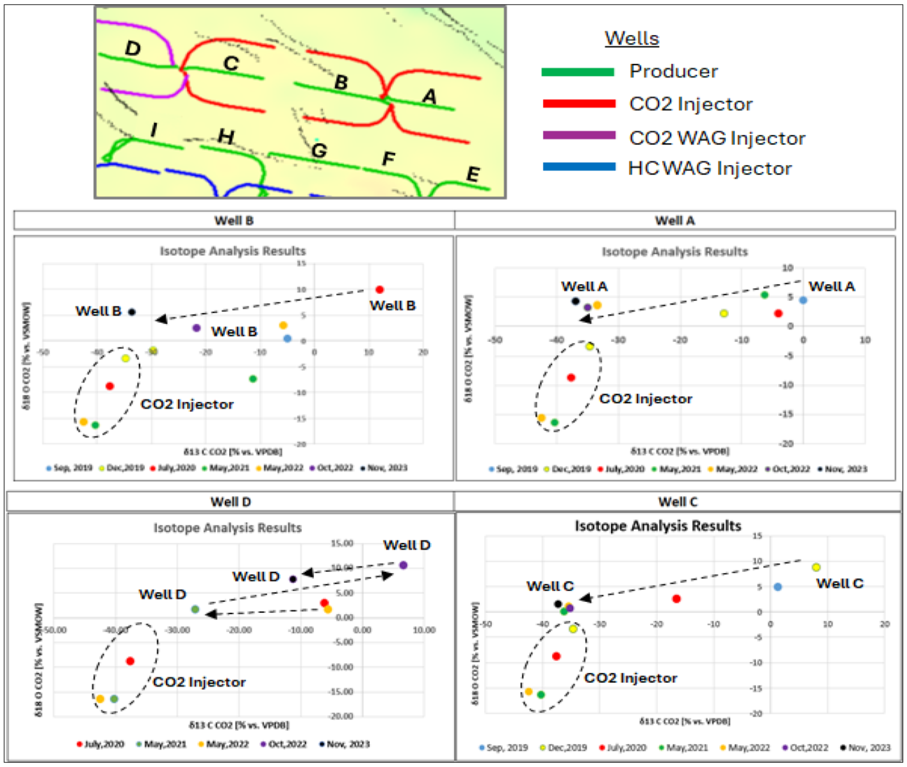

Figure 7—Four producer wells partially or fully affected by the CO_2 injection (Koksalan et al., 2024).

Throughout the CO_2 injection and monitoring program, both the $\delta^{13}C$ isotope composition of CO_2 and its molar concentration have been systematically measured to provide complementary datasets for evaluating reservoir response. Figure 8 presents the results of CO_2 molar concentration analyses for a selection of producer wells monitored over the study period (Koksalan et al., 2024). The data indicate that none of the monitored wells display any discernible impact from CO_2 injection, with measured CO_2 molar concentrations remaining stable in the range of 5–7%, values consistent with their baseline fluid properties prior to injection (Koksalan et al., 2024).

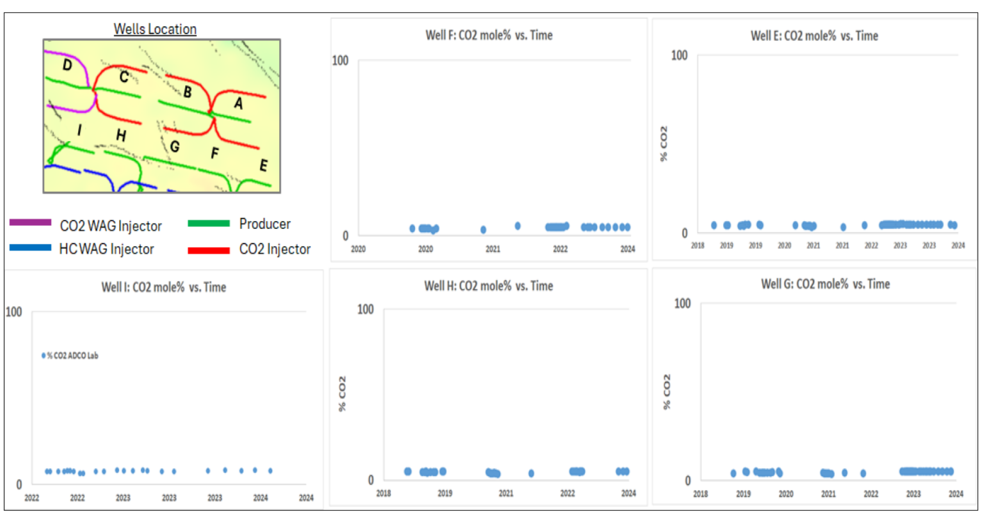

Figure 8—Producer wells not affected by the CO_2 injection (Koksalan et al., 2024).

CO_2 molar concentration measurements were conducted at two independent laboratories, both of which reported identical results. This was done to ensure data reliability and eliminate analytical bias. This duallab verification strengthens confidence in the dataset and confirms the robustness of the findings. The close alignment between the δ[13]C isotope data and the molar concentration results is particularly significant, as both independent methods corroborate the conclusion that injected CO_2 has not yet migrated to these wells. This consistency demonstrates the value of combining isotopic and compositional analyses in CO_2-EOR monitoring programs, providing a strong, cross-validated framework for tracking CO_2 movement and assessing flood progression (Koksalan et al., 2024).

The figure below presents CO_2 molar concentration profiles for four wells, three of which—Wells A, B, and C—show clear evidence of CO_2 breakthrough (Koksalan et al., 2024). To ensure robust data quality and eliminate analytical bias, CO_2 molar concentrations were measured using three independent sources: in-house laboratory, a contractor laboratory, and an in-field CO_2 analyzer. The alignment of these measurements provides strong quality control assurance across the monitoring dataset. The timing of breakthrough varies among the wells, reflecting reservoir heterogeneity: Well A registered CO_2 breakthrough in July 2020, Well B in October 2021, and Well C in June 2022, demonstrating how local geology and flow pathways influence the timing of CO_2 migration (Koksalan et al., 2024).

A particularly noteworthy observation is that δ[13]C isotope measurements consistently detect breakthrough earlier than CO_2 molar concentration data (Koksalan et al., 2024). For instance, in Well B, δ[13]C isotopes began indicating CO_2 breakthrough as early as May 2020, more than two years before changes in CO_2 molar concentration were observed in June–July 2022. This finding highlights the greater sensitivity of isotope monitoring for early detection of CO_2 migration within the reservoir (Koksalan et al., 2024).

Additionally, isotope data from Well D suggest it is already being influenced by injected CO_2, while molar concentration measurements show no discernible impact. This discrepancy underscores the ability of isotopic analysis to detect subtle CO_2 movement that compositional measurements may miss. Taken together, these results reinforce the value of integrating isotopic monitoring with molar concentration measurements and support the conclusion that Water-Alternating-Gas (WAG) injection is a more efficient and controlled EOR process compared to continuous CO_2 injection, enabling earlier detection and improved management of CO_2 flood performance (Koksalan et al., 2024).

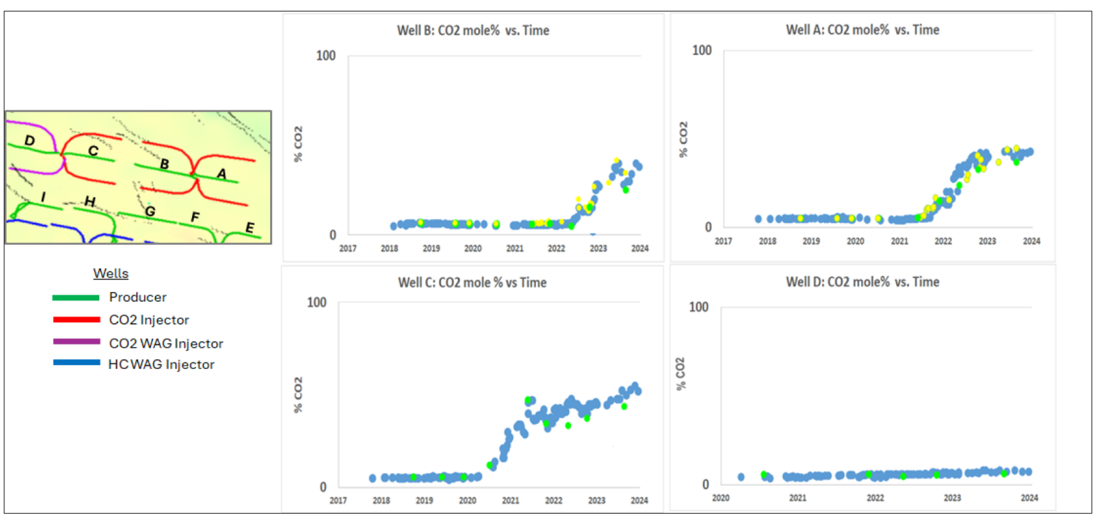

Figure 9—Three producer wells affected by the CO_2 injection as seen by CO_2 molar concentration (Koksalan et al., 2024).

CO_2 Tracers and Model Calibration

For the CO_2 clusters, inter-well tracer campaigns are in place to monitor the CO_2 advancement. Rapid increase in CO_2 mole fraction in the production wells has been observed in some producers and is closely monitored. Without any model adjustments, no CO_2 breakthrough is observed, indicating the need to reevaluate geological understanding and the modelling perspectives (refer to the figure below).

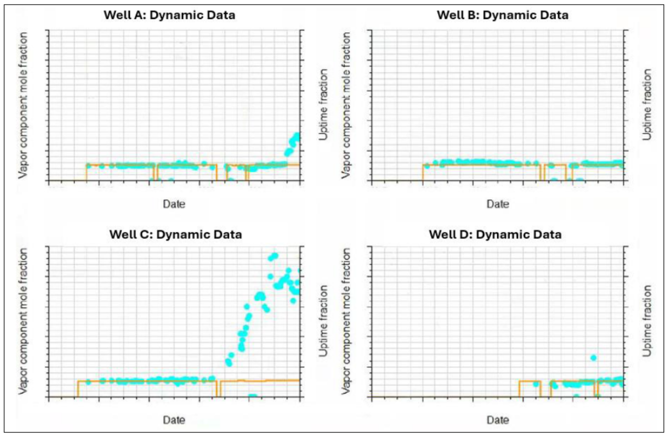

Figure 10—CO_2 breakthrough profiles before model calibration

CO_2 is mostly flowing in the upper layer of the main zone. A approach is utilized to match the CO_2 breakthrough by increasing the permeability contrast in the best rock type to facilitate the gas flow in the upper layer. Any modification is restricted to the upper most layer of the zone. This modification emulates possible fractures in the model which will be investigated in a separate project scope. Sub-seismic faults are also investigated and one fault near Well C is modelled as very conductive.

The model calibration process has demonstrated the robustness of our predictive model. By adjusting the model parameters and validating against measured data, a fair degree of alignment is achieved between the predicted and observed outcomes. This calibration not only enhances the model's reliability but also ensures its applicability across various scenarios.

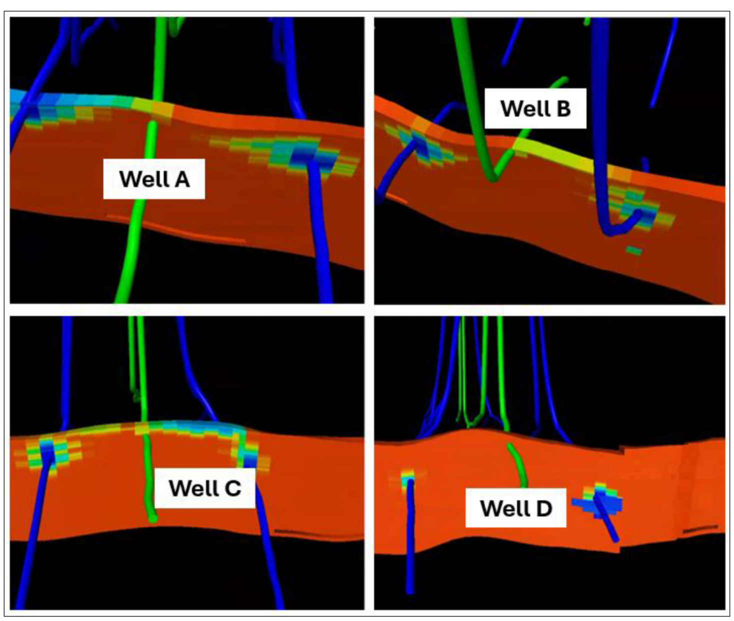

Figure 11—CO_2 distribution after model calibration.

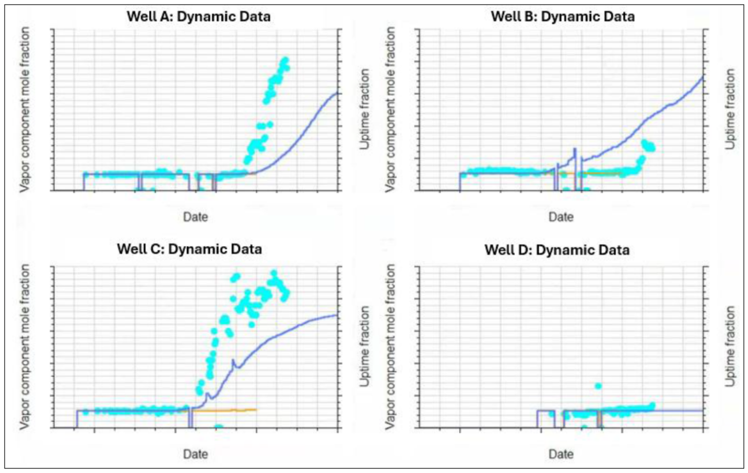

Figure 12—CO_2 production profile before and after model calibration.

Discussion

While the integrated monitoring and modeling approach has provided valuable insights into CO_2 movement and sweep behavior, several challenges remain in fully understanding and optimizing CO_2-EOR performance in the field. One of the primary difficulties is data uncertainty and representativeness; sampling for CO_2 composition and isotopes in production wells is inherently limited by operational constraints and well accessibility, leading to potential gaps in the dataset (Pearce et al. 2013). These gaps can complicate trend analysis and delay the detection of early CO_2 breakthrough events, which are critical for proactive reservoir management. In addition, the isotopic signatures of CO_2 can be influenced by geochemical interactions between the injected gas, formation water, and reservoir rock, including processes like CO_2 dissolution or mineral precipitation (Benson and Cole 2008). These reactions can alter the original isotopic fingerprint of the injected gas, creating uncertainties when interpreting monitoring results.

Another significant challenge stems from reservoir heterogeneity. Abu Dhabi's carbonate reservoirs, like many others in the region, exhibit complex stratigraphy, with layered permeability and localized fractures that can result in preferential flow pathways (Al-Mutairi and Graves 2008). This heterogeneity can cause uneven sweep and localized CO_2 channeling, which may not be fully captured by routine monitoring. Additionally, gravity segregation can lead to vertical CO_2 migration into higher layers, bypassing oil in lower zones and reducing displacement efficiency (Tzimas and Georgakaki 2005). Addressing this behavior in reservoir models requires constant recalibration and fine-tuning, as even small discrepancies in permeability contrast or fracture representation can have a large impact on predictive accuracy.

From an operational perspective, maintaining consistent, high-quality data streams from tracers, CO_2 analyzers, and isotopic measurements is logistically demanding and costly. Frequent sampling, laboratory analyses, and equipment maintenance require significant investment and coordination (Han et al. 2010). In remote or offshore environments, these challenges are amplified by accessibility issues and safety considerations. Integrating diverse datasets also presents difficulties: inconsistencies between tracer results, isotope analyses, and pressure-production data may arise, requiring careful reconciliation to avoid model miscalibration (Almulla and Al Hosani 2021).

Finally, there is the broader issue of scaling pilot findings to field-wide application. While monitoring methods may work effectively in controlled pilot areas, replicating the same level of detail across an entire field is often impractical. This raises questions about the balance between monitoring resolution, cost, and operational benefit, when operators aim to expand CO_2-EOR as part of a long-term carbon management strategy. Despite these challenges, advances in real-time CO_2 sensors, improved tracer technologies, and machine learning–based model calibration hold promise for enhancing monitoring precision and decision-making in future CO_2-EOR projects.

Conclusions

This study demonstrates the value of integrating CO_2 isotopic data, molar concentrations, tracer measurements, and calibrated reservoir models for monitoring CO_2-EOR in an Onshore field. Key conclusions include:

- Isotopic monitoring offers earlier breakthrough detection than molar concentration data, providing a critical tool for proactive management.
- Reservoir heterogeneity and gravity segregation continue to drive uneven sweep and require advanced modeling and mitigation strategies.
- Transitioning from continuous CO_2 injection to WAG is a viable mitigation strategy for managing breakthrough and improving sweep efficiency.
- Integrated monitoring frameworks are essential for optimizing CO_2-EOR performance, and with continued innovation, they can support scalable, cost-effective CO_2 management in Abu Dhabi and beyond.
- Model calibration process to improve the match of CO_2 breakthrough has demonstrated the robustness of the predictive model. By adjusting the model parameters and validating against measured data, a fair degree of alignment is achieved between the predicted and observed outcomes. This calibration not only enhances the model's reliability but also ensures its applicability across various scenarios.

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